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Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2022-2023
FULL MARKS: 75

Math 4543: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. a) i. Define numerical methods. Briefly explain why we need them.

2
(CO2)
(PO1)

Solution:

Numerical methods are techniques to approximate mathematical processes.

The application of numerical methods is an important step in solving engineering problems. We need to resort to finding approximations because there are some problems that cannot be solved analytically or have intractable analytical solutions.

Rubric:

- 1 point for a correct definition.
- 1 point for a valid explanation.

- ii. How do numerical approaches differ from analytical approaches to solving a problem?

2
(CO2)
(PO1)

Solution:

Any 2 valid differences will suffice. Some differences can be written as follows —

Criteria	Numerical Methods	Analytical Methods
Type of yielded solution	Approximate solution	Exact solution
Working Procedure	Keep iterating until the obtained solution is a “good enough” approximation.	Perform symbolic manipulations to arrive at an equation that yields the exact solution.
Example	Solving $ax^2 + bx + c = 0$ using the quadratic equation $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	Solving $ax^2 + bx + c = 0$ using any root-finding algorithm e.g. the Bisection method.

Rubric:

- 1 point for each valid difference.

- b) i. Suppose, R is an arbitrary number. Prove that the *Newton-Raphson* formula for finding the inverse of the square root of R , which is mathematically denoted as $\frac{1}{\sqrt{R}}$, is as shown in Equation-1. 5
(CO2)
(PO1)

$$x_{i+1} = \frac{1}{2} \left(x_i + \frac{1}{Rx_i} \right) \quad (1)$$

Solution:

We know, the Newton-Raphson method formula for solving $f(x) = 0$ is,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Suppose,

$$\begin{aligned} x &= \frac{1}{\sqrt{R}} \\ \Rightarrow x^2 &= \frac{1}{R} \\ \Rightarrow x^2 - \frac{1}{R} &= 0 \end{aligned}$$

We take $f(x) = x^2 - \frac{1}{R}$ and so $f'(x) = 2x$.

Thus,

$$\begin{aligned} x_{i+1} &= x_i - \frac{x_i^2 - \frac{1}{R}}{2x_i} \\ &= x_i - \left(\frac{Rx_i^2 - 1}{2Rx_i} \right) \\ &= x_i - \left(\frac{x_i}{2} - \frac{1}{2Rx_i} \right) \\ &= x_i - \frac{x_i}{2} + \frac{1}{2Rx_i} \\ &= \frac{x_i}{2} + \frac{1}{2Rx_i} \\ &= \frac{1}{2} \left(x_i + \frac{1}{Rx_i} \right) \quad \square \text{ Q.E.D.} \end{aligned}$$

Rubric:

- 1 point for using the Newton-Raphson approximation formula.
- 1 point for defining $f(x)$ and $f'(x)$ properly.
- 3 points for correct derivation.

- ii. What is the advantage of formulating $\frac{1}{\sqrt{R}}$ like Equation-1 using the *Newton-Raphson* method?

2
(CO3)
(PO1)

Solution:

The instructions for the division operator (`fdiv`) and the square root operator (`fsqrt`) comparatively require a lot more clock cycles. The Newton-Raphson formulation of $\frac{1}{\sqrt{R}}$ can be seen as an efficient substitute for these costly mathematical operations.

Rubric:

- 1 point for mentioning that division is costly.
- 1 point for mentioning that square root is costly.

- c) The *Silver Ratio*, often denoted by the Greek letter δ (delta), is the ratio of two quantities a and b with $a > b > 0$, if the ratio $\frac{2a+b}{a} = \frac{a}{b}$. The constant δ is equal to the positive root of the quadratic equation $x^2 = 2x + 1$, which is

14
(CO2)
(PO1)

$$x_{root} = \delta = 1 + \sqrt{2} = 2.41421356237 \dots$$

Apply the *Regula Falsi* method, also known as the *False Position* method, to estimate the value of the silver ratio δ . The initial upper and lower guesses are $x_l = 2$ and $x_u = 3$ respectively. Demonstrating the step-by-step mathematical procedure, conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m) at the end of each iteration. Draw Table-1 on your answer script and fill it out after performing the necessary calculations.

Table 1: The relevant values obtained in the answer of Question-1(c).

Iteration	x_l	x_u	x_r	$ \epsilon_a \%$	m	$f(x_r)$
1	2	3		—	—	
2						
3						
4						

Solution:

After performing the series of calculations, we end up with the following table.

Iteration	x_l	x_u	x_r	$ \epsilon_a \%$	m	$f(x_r)$
1	2	3	2.333	—	—	−0.222
2	2.333	3	2.4	2.78	1	−0.04
3	2.4	3	2.4118	0.49	2	−0.0069
4	2.4118	3	2.4138	0.08	2	−0.0002

Rubric:

- 0.5 point for each correct answer.
- 1 point for demonstrating the step-by-step procedure in each iteration.

2. a) Define *Truncation Error*. What are the different avenues through which *Truncation Error* can be introduced? Explain with proper examples.

5
(CO3)
(PO1)

Solution:

Truncation Error is the error caused by approximating the mathematical process or the modeling technique. Some of the ways it can be introduced are —

1. **Chopping off terms in a series:** If we consider only the first 3 terms of the Maclaurin series of e^x , we get $e^x \approx \left(1 + x + \frac{x^2}{2!}\right)$. And so the truncation error is,

$$\text{Truncation Error} = \text{Exact Value} - \text{Approximate Value}$$

$$= e^x - \left(1 + x + \frac{x^2}{2!}\right)$$

2. **Using a finite step size instead of an infinitesimal value:** To find the derivative of a function $f(x)$ using a finite step size h instead of an infinitesimally small h , the truncation error is,

$$\text{Truncation Error} = \text{Exact Value} - \text{Approximate Value}$$

$$= \left(\lim_{h \Rightarrow 0} \frac{f(x+h) - f(x)}{h}\right) - \left(\frac{f(x+h) - f(x)}{h}\right)$$

3. **Approximating area under curve with rectangles:** To find the integral value of a function $f(x)$ within given bounds a and b using n rectangles of width Δx_i and height $f(x_i)$, the truncation error is,

$$\text{Truncation Error} = \text{Exact Value} - \text{Approximate Value}$$

$$= \int_a^b f(x)dx - \sum_{i=0}^n f(x_i) \cdot \Delta x_i$$

Rubric:

- 1 point for a correct definition.
- 2 points for each valid example. Any 2 valid examples will suffice.

b) The Taylor series for a function $f(x)$ is —

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of

$$f(x) = \sec(x)$$

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(\sec(x)) = \sec(x)\tan(x)$, and $\frac{d}{dx}(u(x)v(x)) = u(x)\frac{d}{dx}(v(x)) + v(x)\frac{d}{dx}(u(x))$. Now, for $f(x) = \sec(x)$, let's calculate the derivatives at the point $x = 0$.

$$f(x) = \sec(x), \quad f(0) = 1$$

$$f'(x) = \sec(x)\tan(x), \quad f'(0) = 0$$

$$f''(x) = \sec^3(x) + \sec(x)\tan^2(x), \quad f''(0) = 1$$

$$f'''(x) = 5\sec^3(x)\tan(x) + \sec(x)\tan^3(x), \quad f'''(0) = 0$$

$$f^{(4)}(x) = 5\sec^5(x) + 18\sec^3(x)\tan^2(x) + \sec(x)\tan^4(x), \quad f^{(4)}(0) = 5$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$f(0+h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots$$

$$\Rightarrow f(h) = 1 + 0 + \frac{h^2}{2!} + 0 + \frac{5h^4}{4!} + \dots$$

$$\Rightarrow f(x) = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \dots$$

$$\Rightarrow \sec(x) = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \dots$$

$$\Rightarrow \sec(x) = 1 + \frac{x^2}{2} + \frac{5x^4}{24} + \dots$$

Rubric:

- 1 point for defining $f(x)$ and finding each i -th order derivative $f^i(x)$.
- 1 point for correct values of $f(0)$ and each $f^i(0)$.
- 5 points for deriving the correct series.

c) The Maclaurin series of the exponential function e^x is —

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

i. Why is the exponential function e^x categorized as a *transcendental function*?

1

Solution:

The exponential function e^x cannot be expressed using a finite sequence of terms or algebraic operations which is why it falls under the umbrella of transcendental functions.

Rubric:

- 1 point for the correct reason.

(CO4)

(PO1)

ii. Using the *Remainder Theorem*, establish the bounds of the truncation error in the representation of $e^{1.69}$ if only the first 5 terms of the series are used.

4

(CO2)

(PO1)

Solution:

If we are using 5 terms of the Taylor series, the remainder is given by ($x = 0$, $h = 1.69$, $n = 4$)

$$R_4(0 + 1.69) = \frac{1.69^{(4+1)}}{(4+1)!} f^{(4+1)}(c) = \frac{1.69^5}{5!} f^5(c) = \frac{1.69^5 e^c}{120}$$

Since,

$$\begin{aligned} x &< c < x + h \\ 0 &< c < 0 + 1.69 \\ 0 &< c < 1.69 \end{aligned}$$

The error bound is between,

$$\begin{aligned} \frac{1.69^5 e^0}{120} &< R_4(1.69) < \frac{1.69^5 e^{1.69}}{120} \\ 0.1148 &< R_4(1.69) < 0.6226 \end{aligned}$$

Rubric:

- 1 point for finding $R_4(1.69)$ in terms of c .
- 1.5 points for correct lower-bound.
- 1.5 points for correct upper-bound.

3. a) Differentiate between *Interpolation* and *Extrapolation*. With suitable real-world examples, briefly write when each of the methods is applicable.

4
(CO3)
(PO1)

Solution:

Any 2 valid conceptual differences will suffice. There should be at least 1 valid real-world example that is used to explain the applicability. Some differences are —

Criteria	Interpolation	Extrapolation
Position of newly estimated points	Within the distribution of the known data points.	Outside the distribution of the known data points.
Likelihood of proper estimation	Has a greater likelihood of obtaining a valid estimate, since the pattern of the data points on either side of the newly estimated point is known.	Has a lesser likelihood of obtaining a valid estimate, since it assumes that the observed trend continues for values of x outside the range of the given data points.
Applicability	• Computer Graphics – For constructing B-splines, smooth surfaces, etc. • Carpentry – For drawing curves on wooden surfaces using different apparatus e.g. the Spline tool. • Digital Image Processing – For increasing the resolution of images. • Electronics – For plotting transfer functions or calibration curves of electronic components e.g. Thermistors.	

Rubric:

- 1 point for each valid conceptual difference.
- 2 points for a valid real-world application.

- b) Suppose, there is an unknown function $f(x)$ and you are given 3 points $(x_0, f(x_0))$, $(x_1, f(x_1))$, and $(x_2, f(x_2))$ through which $f(x)$ passes. Now, in order to approximate $f(x)$, you decide to use the *Newton's Divided Difference* method of interpolation and obtain a *quadratic interpolant* $f_2(x)$. Derive this 2nd order Newton's Divided Difference polynomial $f_2(x)$. 8
(CO4)
(PO1)

Solution:



Just [click here](#).

Rubric:

- 1 point for writing the quadratic NDD interpolant $f_2(x)$ in the proper form.
- 1 point for correct derivation of b_0 .
- 1 point for correct derivation of b_1 .
- 4 points for correct derivation of b_2 .
- 1 point for writing the correct expanded form of the quadratic NDD interpolant $f_2(x)$.

- c) i. Briefly explain the necessity of *Spline Interpolation*. 2
(CO3)
(PO1)

Solution:

Other interpolation methods become susceptible to the Runge phenomenon when the order n of the interpolant is gradually increased (due to the manifestation of an unhinged oscillatory behavior at either end of the interpolant). We need Spline interpolation since it is immune to this Runge phenomenon.

Rubric:

- 2 points for mentioning its immunity to the Runge phenomenon.

- ii. Suppose you want to perform Spline interpolation on $n + 1$ data points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $\dots$, and $(x_n, f(x_n))$ using *Cubic splines* of the form $f(x) = ax^3 + bx^2 + cx + d$. How many unknowns and how many equations would you have to deal with? Mention how you would obtain those equations and derive them. 5
(CO4)
(PO1)

Solution:

Each cubic spline has 4 unknowns. So, for n such splines, we have a total of $4n$ unknowns. To obtain these $4n$ unknowns, we need $4n$ equations.

Here, we have $n + 1$ given data points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $\dots$, and $(x_n, f(x_n))$. The n cubic splines that go through consecutive points and form the interpolant $f(x)$ are,

$$f(x) = \begin{cases} a_1x^3 + b_1x^2 + c_1x + d_1; & \text{for } x_0 \leq x \leq x_1 \\ a_2x^3 + b_2x^2 + c_2x + d_2; & \text{for } x_1 \leq x \leq x_2 \\ \vdots & \vdots \\ a_nx^3 + b_nx^2 + c_nx + d_n; & \text{for } x_{n-1} \leq x \leq x_n \end{cases}$$

Now, let's derive the $4n$ equations. Since the cubic splines pass through consecutive points, we get the following $2n$ equations.

$$\begin{aligned}
 a_1x_0^3 + b_1x_0^2 + c_1x_0 + d_1 &= f(x_0) \\
 a_1x_1^3 + b_1x_1^2 + c_1x_1 + d_1 &= f(x_1) \\
 \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 a_1x_{i-1}^3 + b_1x_{i-1}^2 + c_1x_{i-1} + d_1 &= f(x_{i-1}) \\
 a_1x_i^3 + b_1x_i^2 + c_1x_i + d_1 &= f(x_i) \\
 \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 a_1x_{n-1}^3 + b_1x_{n-1}^2 + c_1x_{n-1} + d_1 &= f(x_{n-1}) \\
 a_1x_n^3 + b_1x_n^2 + c_1x_n + d_1 &= f(x_n)
 \end{aligned}$$

Now, find out the 1st order derivatives of the consecutive cubic splines at the $n - 1$ interior points, since they are continuous at those points. We obtain another $n - 1$ equations in this way.

$$\begin{aligned}
 3a_1x_1^2 + 2b_1x_1 + c_1 &= 3a_2x_1^2 + 2b_2x_1 + c_2 \\
 \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 3a_{n-1}x_{n-1}^2 + 2b_{n-1}x_{n-1} + c_{n-1} &= 3a_nx_{n-1}^2 + 2b_nx_{n-1} + c_n
 \end{aligned}$$

Now, find out the 2nd order derivatives of the consecutive cubic splines at the $n - 1$ interior points, since they are continuous at those points. We obtain yet another $n - 1$ equations in this way.

$$\begin{aligned}
 6a_1x_1 + 2b_1 &= 6a_2x_1 + 2b_2 \\
 \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 6a_{n-1}x_{n-1} + 2b_{n-1} &= 6a_nx_{n-1} + 2b_n
 \end{aligned}$$

Now, we require 2 more equations. For these, we consider $a_1 = 0$ and $a_{n-1} = 0$. So, in aggregate, we have $2n + (n - 1) + (n - 1) + 2 = 4n$ equations for the $4n$ unknowns.

Rubric:

- 1 point for the correct number of unknowns and equations.
- 1 point for the initial $2n$ equations.
- 1 point for the $n - 1$ equations using 1st order derivatives.
- 1 point for the $n - 1$ equations using 2nd order derivatives.
- 1 point for the last 2 equations.

- d) A thermistor, also known as thermal resistor, is a semiconductor type of resistor whose resistance is strongly dependent on temperature. To measure temperature using a thermistor, the manufacturers provide users with a Temperature (T) vs. Resistance (R) calibration curve. If we measure the resistance of the thermistor using an Ohmmeter, then we can refer to the aforementioned curve and determine the corresponding temperature value. Figure-1 and Table-2 portray multiple recorded observations involving a thermistor.

6
(CO2)
(PO1)

Table 2: Temperature T of a thermistor as a function of its resistance R for Question-3(d).

Resistance, R (Ω)	Temperature, T ($^{\circ}C$)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

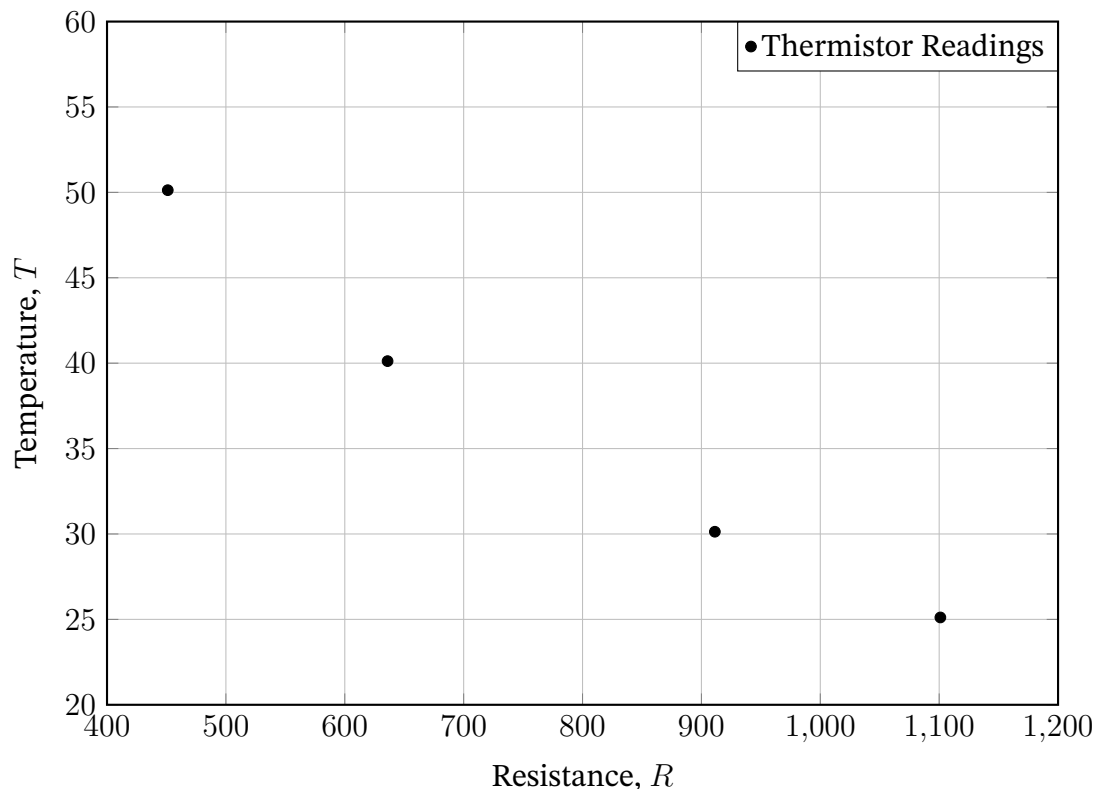


Figure 1: Temperature T vs. Resistance R for Question-3(d).

Determine the value of the temperature T when the thermistor resistance is $R = 754.8\Omega$ using the *Lagrangian method* of interpolation and a second order polynomial.

Solution:

The 2nd order Lagrangian interpolant is —

$$\begin{aligned}
 T(R) &= \sum_{i=0}^2 L_i(R)T(R_i) \\
 &= L_0(R)T(R_0) + L_1(R)T(R_1) + L_2(R)T(R_2)
 \end{aligned}$$

Given, $R = 754.8\Omega$, and the 3 nearest data points that bracket it are, $R_0 = 911.3$, $R_1 = 636$, and $R_2 = 451.1$. The corresponding T values are $T(R_0) = 30.131$, $T(R_1) = 40.12$, and $T(R_2) = 50.128$ respectively.

Now, let's calculate the Lagrangian weights.

$$L_0(R) = \prod_{\substack{j=0 \\ j \neq 0}}^2 \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 636}{911.3 - 636} \right) \times \left(\frac{754.8 - 451.1}{911.3 - 451.1} \right) = 0.284$$

$$L_1(R) = \prod_{\substack{j=0 \\ j \neq 1}}^2 \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 911.3}{636 - 911.3} \right) \times \left(\frac{754.8 - 451.1}{636 - 451.1} \right) = 0.933$$

$$L_2(R) = \prod_{\substack{j=0 \\ j \neq 2}}^2 \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 911.3}{451.1 - 911.3} \right) \times \left(\frac{754.8 - 636}{451.1 - 636} \right) = -0.218$$

Now that we have the weights, we can obtain the value of $T(754.8)$.

$$\begin{aligned} T(754.8) &= 0.284 \times T(911.3) + 0.933 \times T(636) + (-0.218) \times T(451.1) \\ &= 0.284 \times 30.131 + 0.933 \times 40.12 + (-0.218) \times 50.128 \\ &\approx 35.088^\circ\text{C} \end{aligned}$$

Rubric:

- 1 point for the correct polynomial representation.
- 1 point for choosing the correct nearest points.
- 1 point for each Lagrangian weight $L_i(R)$ value.
- 1 point for the correct answer.